SAMSUNG

Samsung Innovation Campus

AI Course

Together for Tomorrow!
Enabling People
Education for Future Generations

SAMSUNG

30-Day Hospital Readmission Risk Prediction for Patients with Diabetes Using Machine Learning

HexaCare (team 13)

Artificial Intelligence Course

Together for Tomorrow!
Enabling People
Education for Future Generations

30-Day Hospital Readmission Risk Prediction for Patients with Diabetes Using Machine Learning

Unit 1. Introduction

1.1 Problem Statement & Objective

1.2 Problem Statement

Unit 2. Data & Solution Development

2.1 Data Acquisition & Understanding

2.2 Exploratory Data Analysis

2.3 Data Preprocessing & Feature Engineering

2.4 Machine Learning Pipeline

Unit 3. Results, Application & Impact

3.1 Key Results

3.2 User Interface

3.3 Testing & Improvements

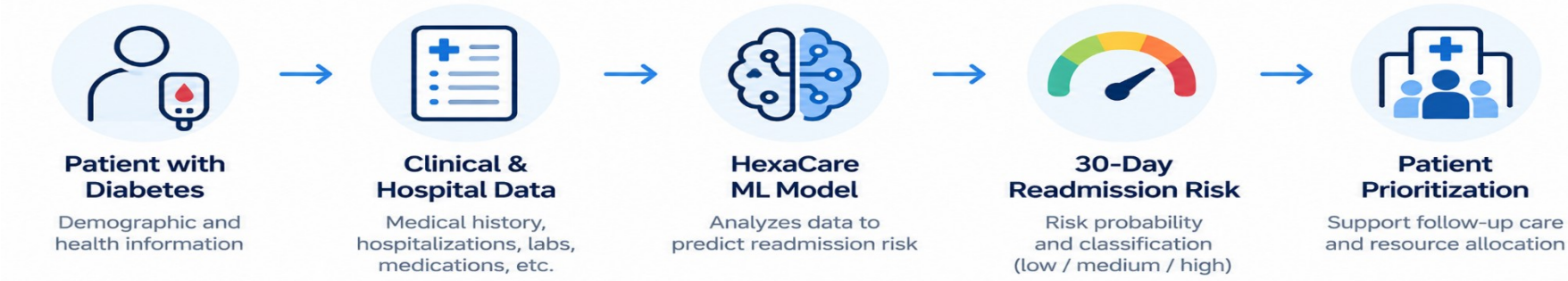
3.4 Application & Future Impact

3.5 Conclusion

1.1. Problem Statement & Objective

- Hospital readmission within 30 days is an important healthcare challenge, particularly among patients with diabetes. HexaCare addresses this challenge by using machine learning to analyze patient demographic, clinical, hospitalization, and healthcare-utilization data to estimate readmission risk and support the early identification and prioritization of high-risk patients.

- Objective
    - To develop a machine learning–based system for predicting 30-day hospital readmission risk among patients with diabetes.



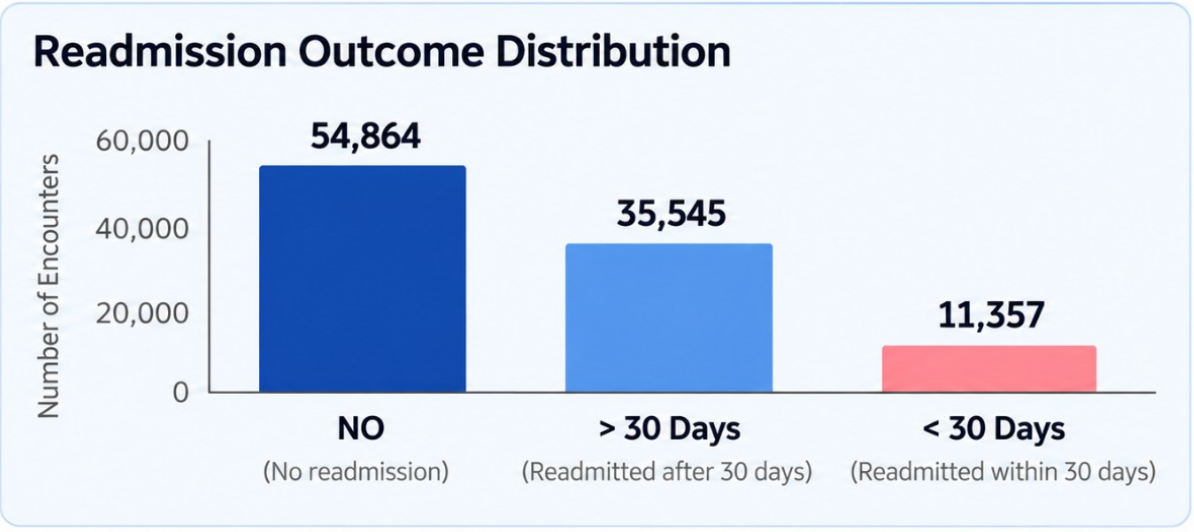
1.2. Problem Statement

- Patients with diabetes may be readmitted to the hospital shortly after discharge, but identifying which patients are at higher risk of 30-day readmission remains challenging. Without early risk identification, healthcare providers may miss opportunities to prioritize follow-up and allocate resources to patients who need closer attention.



Dataset Overview

- **Dataset**: Diabetes 130-US Hospitals
- **Population**: Patients with diabetes across U.S. hospitals
- **Dataset Size**: 101,766 encounters from 71,518 unique patients
- **Variables:** 50 clinical and administrative features
- **Prediction Target**: Readmission within 30 days after discharge



11.16%

Positive Class

Only 11.16% of encounters were readmitted within 30 days.

This indicates a class imbalance, which we address during model development.

Key EDA Insights

What Did We Observe?

- **Higher Prior Healthcare Utilization**
  Readmitted patients showed greater previous inpatient utilization.
- **Longer Hospitalization**
  Readmitted patients had a slightly longer average hospital stay.
- **Higher Medication Use**
  Readmitted patients received slightly more medications on average.



KEY TAKEAWAY Previous inpatient utilization showed the clearest descriptive separation between readmitted and non-readmitted patients.

Preparing the Data for Modeling

1 Data Cleaning

- Excluded ineligible death/hospice encounters.
- Handled missing values and data quality issues.
- Removed irrelevant or high-missing features.

2 Feature Engineering

- Grouped high-cardinality diagnosis codes into clinically meaningful categories.
- Replaced raw diagnosis codes with grouped diagnosis features for modeling.

3 Feature Selection

- Selected **37** modeling features.
- **8** Numerical features.
- **29** Categorical features.
- Focused on clinically relevant and non-redundant features.

4 Patient-Level Split

- Train: **79,541** encounters
- Test: **19,802** encounters
- Split at the patient level to avoid information leakage.

0 Patient Overlap

5 Leakage-Safe Preprocessing

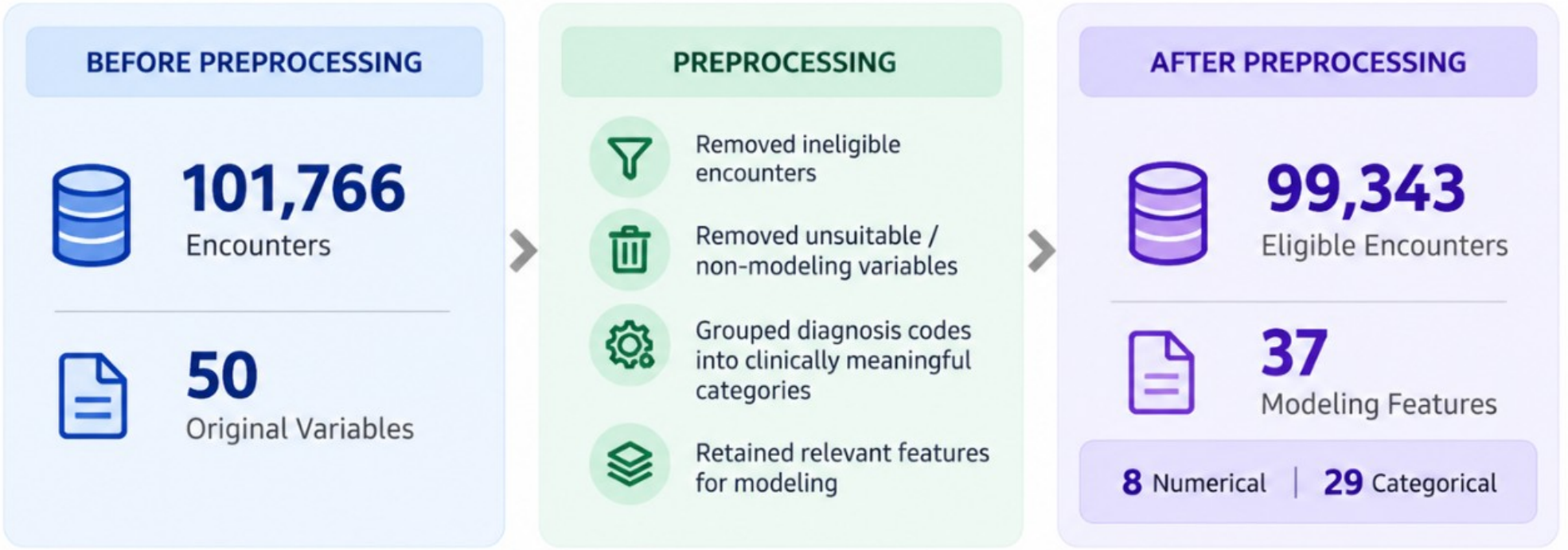
- Scaling and One-Hot Encoding fitted inside cross-validation.
- Ensured no information from test set was used during training.
- All preprocessing steps integrated into the ML pipeline.

WHY IT MATTERS

Preprocessing was performed within the ML pipeline to prevent data leakage and ensure reliable model evaluation.

From Raw Data to Model-Ready Features

Before vs. After

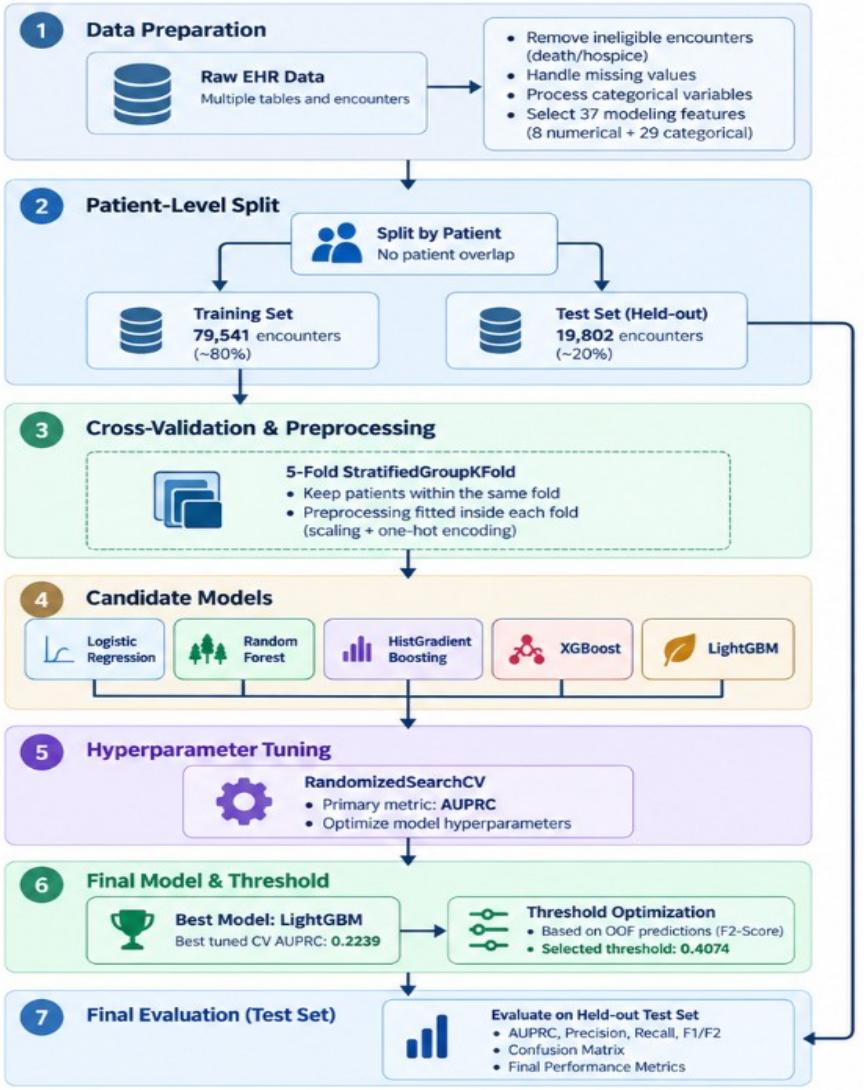


WHY 37 FEATURES?

The final 37 features resulted from removing identifiers, high-missingness, constant, and extremely rare variables, while replacing high-cardinality diagnosis codes with clinically grouped categories.

From Data to Readmission Risk

- **Data Preparation** → Cleaned data and selected **37 modeling features**
  (8 numerical + 29 categorical)
- **Patient-Level Split** → Training and test sets separated by patient
  (**0 patient overlap**)
- **Cross-Validation** → 5-Fold StratifiedGroupKFold with preprocessing performed inside each fold
- **Model Development** → Logistic Regression, Random Forest, HistGradientBoosting, XGBoost, and LightGBM
- **Optimization** → Hyperparameter tuning using RandomizedSearchCV with **AUPRC** as the primary metric
- **Final Model & Threshold** → LightGBM + optimized threshold (**0.4074**)



3.1 Key Results

Model Comparison & Selection

What is AUPRC?

AUPRC (Area Under the Precision-Recall Curve) measures the trade-off between **Precision and Recall** across different thresholds, summarizing the model's ability to identify positive cases (readmissions).

Why AUPRC instead of Accuracy?

- Only **~11%** of encounters were **positive** (readmitted), so the dataset is highly imbalanced.
- Accuracy can be misleading in imbalanced data.
- For example, predicting all cases as "Not Readmitted" could achieve **~89% accuracy** while detecting **0 readmissions**.

Model Selection

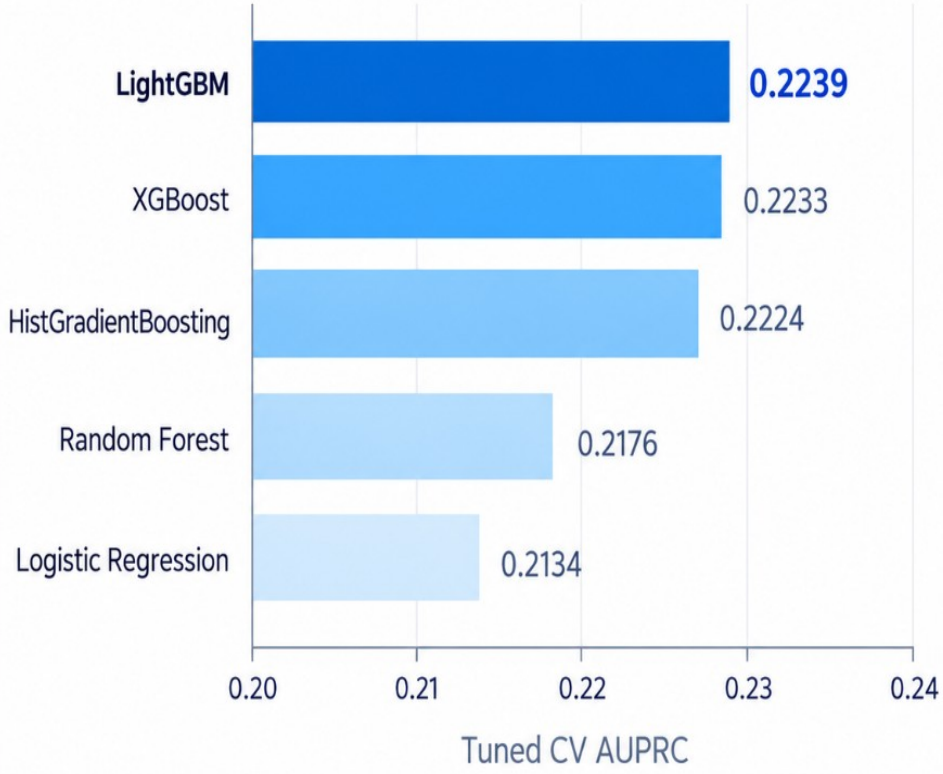
- Five machine learning models were evaluated using patient-grouped 5-fold cross-validation.
- **Higher AUPRC is better ↑.**
- **LightGBM** achieved the highest tuned cross-validated AUPRC **(0.2239)** and was selected as the final model.

Best Model: **LightGBM**
CV AUPRC: **0.2239**

Model Comparison

Tuned Cross-Validated AUPRC (5-Fold)

Higher AUPRC = Better ↑



Final Model Performance & Threshold Optimization

The final LightGBM model was evaluated on the untouched held-out test set. The decision threshold was optimized using out-of-fold training predictions to prioritize recall.

- Final Test Performance → The model achieved ROC-AUC = 0.6677 and AUPRC = 0.2297.
- Threshold Optimization → The decision threshold was reduced from 0.50 to 0.4074 using the F2-score, prioritizing the identification of high-risk patients.
- Recall Improvement → Recall increased from 56.7% to 77.3%, while false negatives decreased from 960 to 503.

KEY IMPACT

The optimized threshold **prioritized recall**, identifying more patients at risk of 30-day readmission while accepting lower precision.

Held-Out Test Performance

Final LightGBM Model on Unseen Test Set

ROC-AUC **0.6677**

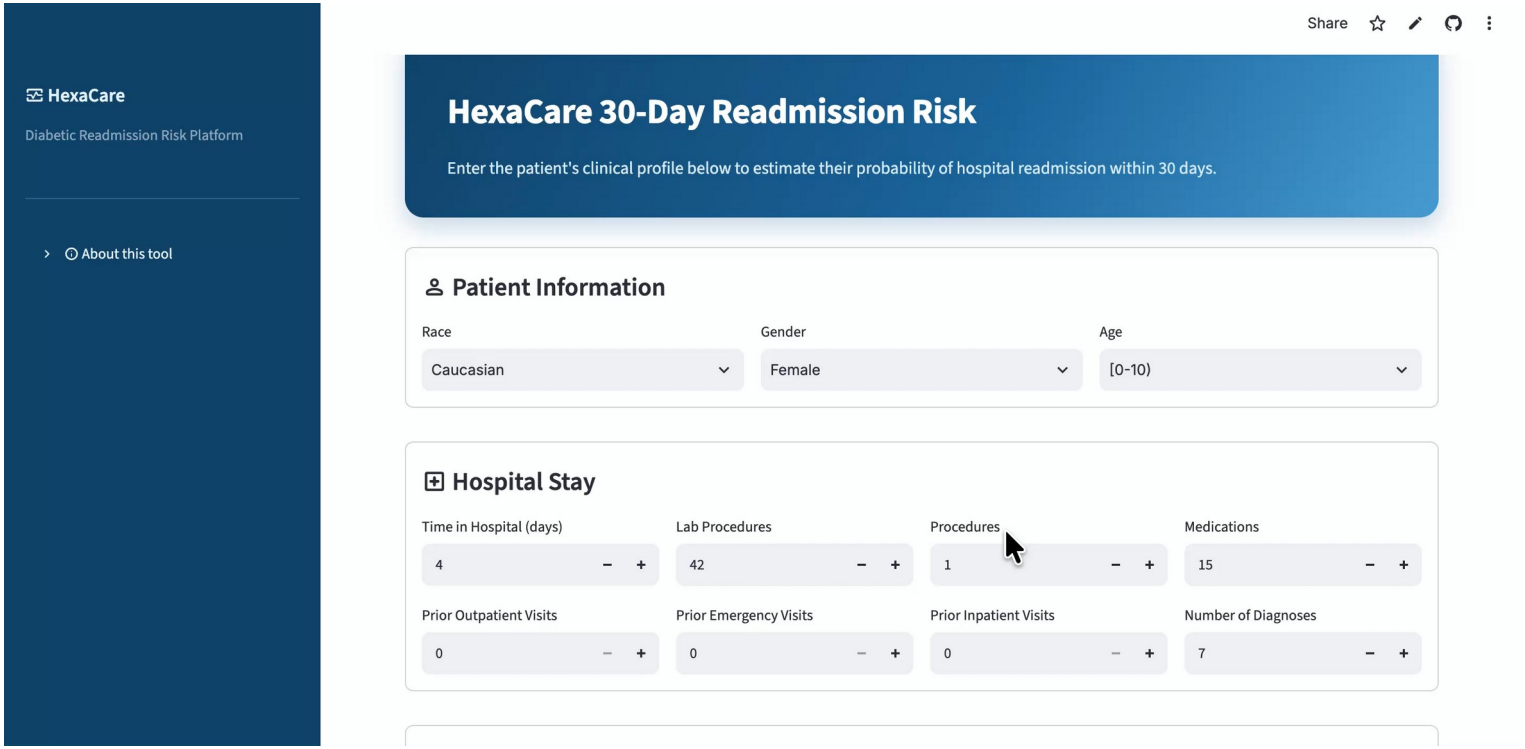
AUPRC **0.2297**

Metric	Default Threshold (0.50)	Optimized Threshold (0.4074)
Threshold	0.50	**0.4074**
Recall (Sensitivity)	56.7%	**77.3%** ↑
False Negatives	960	**503** ↓
True Positives	1,256	**1,713** ↑
Precision (PPV)	17.8%	14.7%

+457 Additional Readmitted Encounters Identified

False negatives reduced by 47.6%

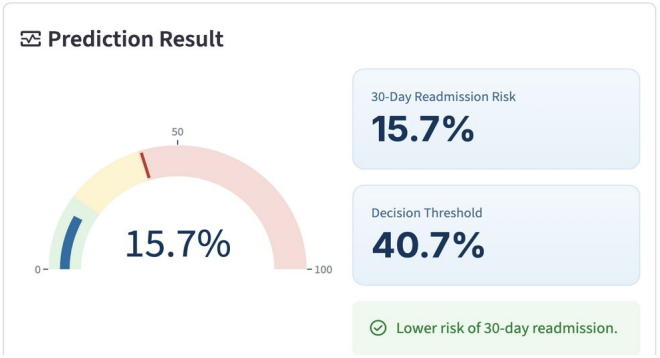
3.2. User Interface



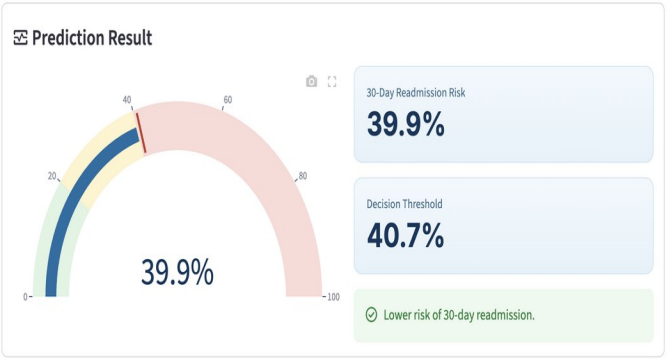
Try the readmission prediction system
[Open ↗]

3.3. Testing and Improvements

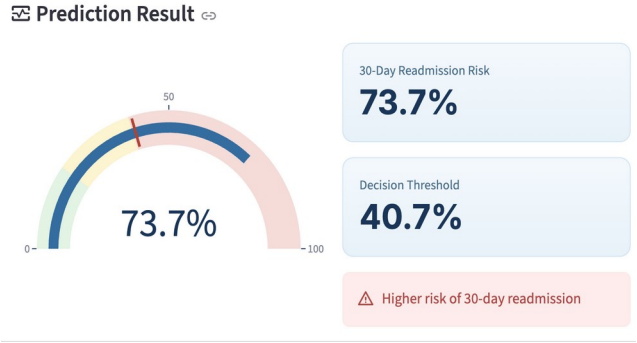
Test 1: Low-Risk Prediction



Test 2: Moderate-Risk Prediction



Test 3: High-Risk Prediction



From Prediction to Healthcare Support

- HexaCare is designed as a risk-screening and decision-support tool to help healthcare providers identify patients who may require closer follow-up after discharge.
- **Early Risk Identification** → Estimates the probability of 30-day readmission before post-discharge follow-up.
- **Patient Prioritization** → Helps identify higher-risk patients who may benefit from closer monitoring and follow-up.
- **Decision Support** → Provides a consistent data-driven risk estimate to complement clinical judgment.



3.5 Conclusion

HexaCare
Towards Better Care

HexaCare: From Data to Actionable Risk

- HexaCare demonstrates how machine learning can be used to support the identification of patients with diabetes who are at higher risk of 30-day hospital readmission.
- A leakage-safe, patient-level modeling pipeline was developed and five machine learning models were evaluated, with **LightGBM selected as the final model**.
- By optimizing the decision threshold, the model achieved **77.3% recall** and reduced false negatives from **960 to 503**, identifying **457 additional readmitted encounters** compared with the default threshold.
- Finally, the model was integrated into a functional user interface that translates model predictions into an accessible risk estimate, demonstrating the potential of HexaCare as a **risk-screening and clinical decision-support tool**.



FINAL MESSAGE

HexaCare demonstrates the potential of machine learning to support earlier identification and prioritization of patients at higher risk of 30-day readmission.

SAMSUNG



©2025 SAMSUNG. All rights reserved.

Samsung Electronics Corporate Citizenship Office holds the copyright of book.

This book is a literary property protected by copyright law so reprint and reproduction without permission are prohibited.
To use this book other than the curriculum of Samsung Innovation Campus or to use the entire or part of this book, you must receive written consent from copyright holder.